

Generation Interconnection System Impact Study Report

For Queue Project Q20-1394

Whirlpool Corporation - Marion Wind Farm 100 kV

March 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1394. The Interconnection Customer proposes a 157.9 MW solar photovoltaic facility interconnecting at the Whirlpool Corporation - Marion Wind Farm 100 kV substation in the Ohio area. The request was received on January 26, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 500 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,198,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Callender to McGill 100 kV	Post Rock to Myrtle Pt 230 kV	210.6	155	5.2
Mortimer to 3Rd Avenue 100 kV	Trego to Galion Generating Station 500 kV	208.6	115	17.4
Napa to Tejon 100 kV	Trego Substation to Beaver 500 kV Line	139.5	131	42.2
Uilk Wind Llc to Napa 100 kV	Trego to Beaver 500 kV line	139.4	131	42.2
Uilk Wind Llc to North Trail 100 kV	Beaver Substation to Trego 500 kV	125.3	154	42.2
Worland Town Substation to Whirlpool Corporation - Marion Wind Farm 100 kV	Uilk Wind Llc to North Trail 100 kV Line	117.7	117	57.5
3Rd Avenue Substation to Bmw Manufacturing Co Substation 100 kV	Trego to Galion Generating Sta 500 kV	113.7	112	9.0
Brookline to Stockton Cogen 100 kV	N Trail Substation to Uilk Wind Llc 100 kV	108.5	123	42.3
N Hills to Stockton Cogen 100 kV	N Trail Substation to Uilk Wind Llc 100 kV	108.4	123	42.3
Tejon to Neligh 100 kV line	Trego to Beaver 500 kV	107.5	151	19.6
Shiras to Telephone 100 kV L	Trego to Beaver 500 kV line	100.5	102	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcgill Substation to Callender 100 kV	Rebuild line with higher-capacity conductor	\$10,648,000	100% (\$10,648,000)
Mortimer to 3Rd Avenue 100 kV line	Rebuild line with higher-capacity conductor	\$6,704,000	100% (\$6,704,000)
Tejon to Napa 100 kV	Rebuild line with higher-capacity conductor	\$8,755,000	91% (\$7,975,000)
Uilk Wind Llc to Napa 100 kV	Rebuild line with higher-capacity conductor	\$1,732,000	91% (\$1,577,000)
Uilk Wind Llc to North Trail 100 kV	Rebuild line with higher-capacity conductor	\$5,089,000	91% (\$4,635,000)
Worland Town to Whirlpool Corporation - Marion Wind Farm 100 kV	Rebuild line with higher-capacity conductor	\$6,084,000	100% (\$6,084,000)
3Rd Avenue Substation to Bmw Manufacturing Co Substation 100 kV	Rebuild line with higher-capacity conductor	\$4,930,000	100% (\$4,930,000)
Brookline to Stockton Cogen 100 kV L	Reconductor line	\$795,000	100% (\$795,000)
Stockton Cogen Substation to N Hills 100 kV line	Reconductor line	\$884,000	100% (\$884,000)
Neligh to Tejon 100 kV L	Reconductor line	\$881,000	79% (\$695,000)
Shiras to Telephone 100 kV L	Reconductor line	\$845,000	32% (\$271,000)
	Total allocated to project		\$45,198,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.